TECH STAR SUMMIT 2025

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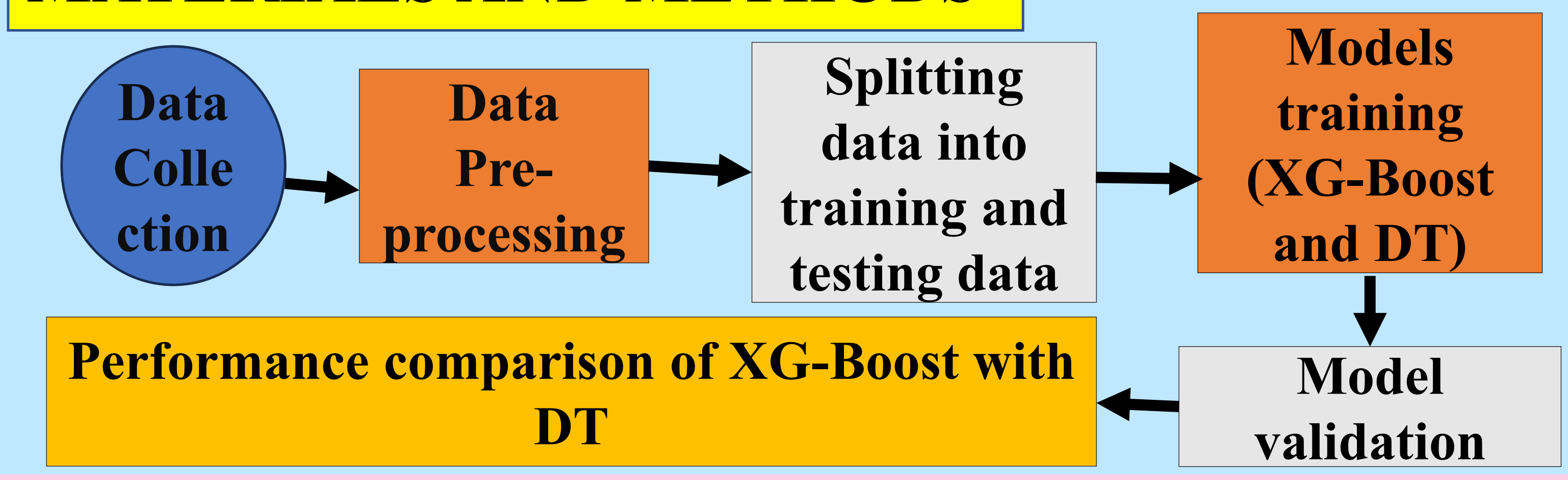
Efficient Prediction of Accuracy in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with Decision Tree Algorithm

INTRODUCTION

- The Research aims to enhance the Accuracy of Inflation Rate Prediction Using Extreme Gradient Boosting Algorithm in comparison with Decision Tree Algorithm
- In the economic and financial sectors, one of the key challenges is the lack of precise and reliable forecasts for inflation rates, which are critical for policy formulation, investment decisions.

Fig1. Inflation Decline

MATERIALS AND METHODS



Data Collection → Data Pre-processing → Splitting data into training and testing data → Models training (XG-Boost and DT) → Model validation → Performance comparison of XG-Boost with DT

RESULTS

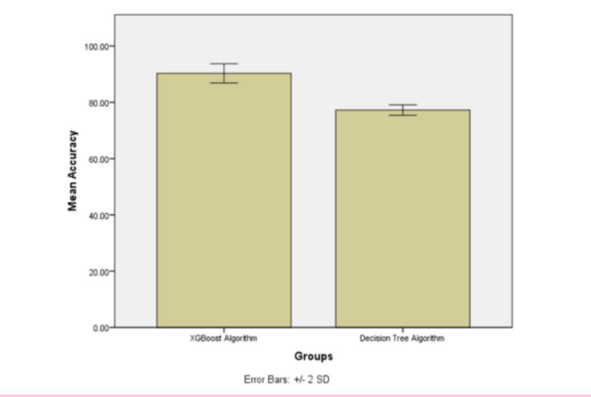


Fig 2 . XGBoost and DT

Table 1 . Standard deviation and mean value of XGBoost and DT

	Algorithm	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	XG-Boost	10	90.115	0.084	0.02659
	DT	10	77.257	0.0048	0.0153

DISCUSSION AND CONCLUSION

- According to this study, the XG-Boost (extreme gradient booster) algorithm performs noticeably better than the Decision Tree method in terms of more accurate inflation rate prediction.
- The Decision Tree algorithm's precision was just 77.25%, but XG-Boost's was 90.11% & p=0.45. The findings show that XG-Boost's capacity to successfully minimize bias and variance leads in a prediction model that is more stable

BIBLIOGRAPHY

- Beer, Christian, Robert Ferstl, and Bernhard Graf. 2025. "Improving Disaggregated Short-Term Food Inflation Forecasts with Webscraped Data." 262. Working Paper.
- Vilenko, Maya. 2025. "BiHRNN -- Bi-Directional Hierarchical Recurrent Neural Network for Inflation Forecasting." http://arxiv.org/abs/2503.01893.
- Huang, Naijing, Yuqing Qi, and Jie Xia. 2024. "China's Inflation Forecasting in a Data-Rich Environment: Based on Machine Learning Algorithms." Applied Economics, March. https://doi.org/10.1080/00036846.2024.2322572.

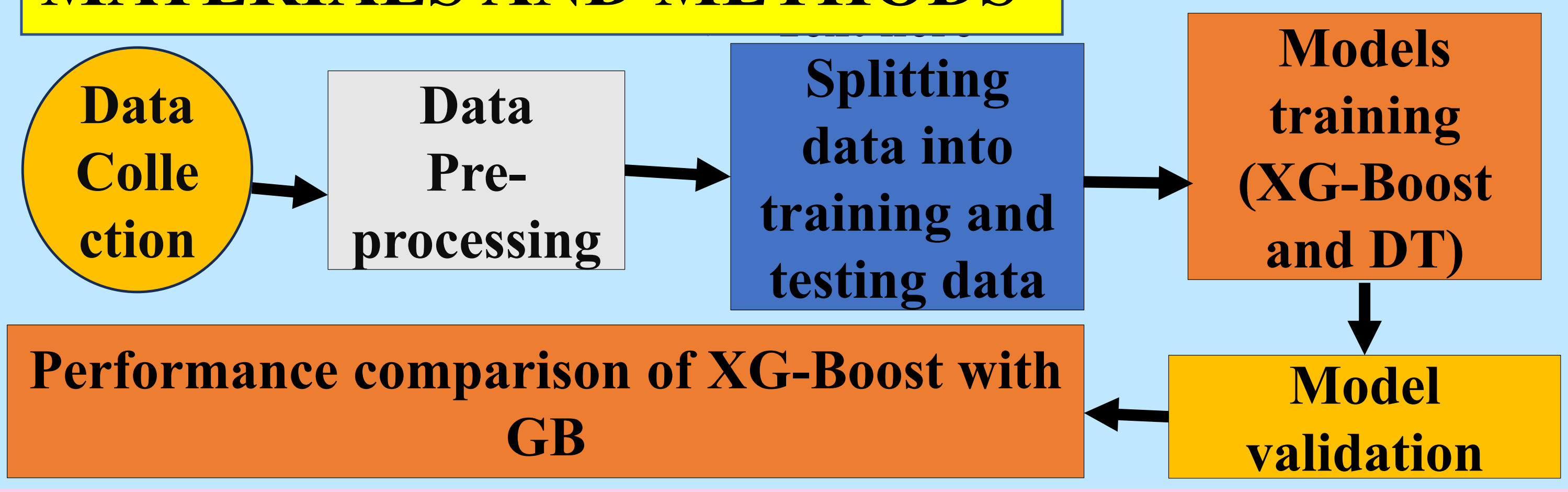
Enhanced Accuracy Prediction in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with Gradient Boosting Algorithm

INTRODUCTION

- The study focuses on improving the precision of inflation rate forecasting by evaluating and comparing the performance of two advanced machine learning techniques: Extreme Gradient Boosting (XG-Boost) and the Gradient Boosting Algorithm (GB).
- Traditional forecasting methods often struggle with nonlinear patterns and high-dimensional data.

Fig1. Rising Prices & Decreasing stability

MATERIALS AND METHODS



Data Collection → Data Pre-processing → Splitting data into training and testing data → Models training (XG-Boost and DT) → Model validation → Performance comparison of XG-Boost with GB

RESULTS

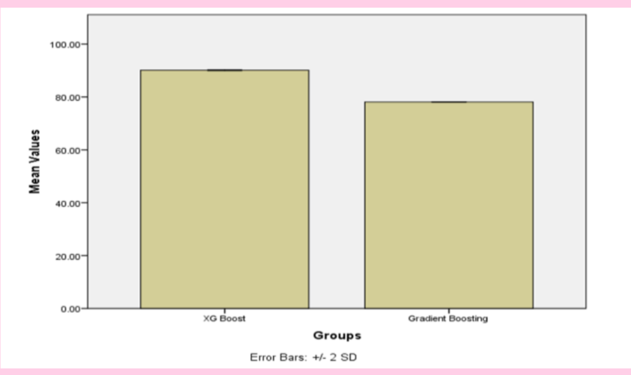


Fig 2 . XGBoost and GB

Table 1 . Standard deviation and mean value of XGBoost and GB

	Algorithm	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	XG-Boost	10	90.115	0.084	0.02659
	GB	10	78.0850	0.0052	0.0167

DISCUSSION AND CONCLUSION

- According to this study, the XG-Boost (extreme gradient booster) algorithm performs better than the Gradient Boosting approach in terms of more accurate inflation rate predictions.
- The accuracy of the Gradient Boosting technique was just 78.08%, whereas XG-Boost's precision was 90.11% & p=0.45. The findings show that XG-Boost offers a more accurate

BIBLIOGRAPHY

- Sun, Lu, Zhixiang Yin, and Lin Lu. 2025. "ISLRWR: A Network Diffusion Algorithm for Drug-Target Interactions Prediction." PloS One 20 (1): e0302281.
- Vilenko, Maya. 2025. "BiHRNN -- Bi-Directional Hierarchical Recurrent Neural Network for Inflation Forecasting." http://arxiv.org/abs/2503.01893.
- Somakumar, Shwetha, Fathima Thashreefa Basheer, Vijayanarayana K, Vani Lakshmi R, Shyamasunder N. Bhat, Gabriel Sunil Rodrigues, Girish Menon R, Elstin Anbu Raj S, and Rajesh V. 2025.

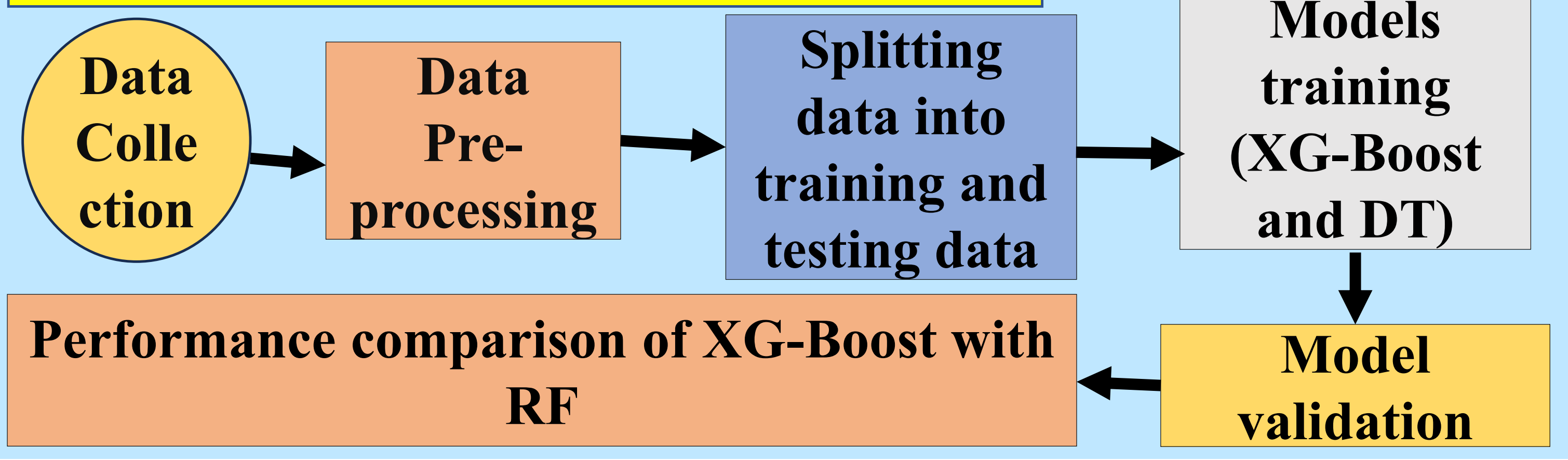
Prediction of Accuracy in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with Random Forest Algorithm

INTRODUCTION

- The research investigates the potential of machine learning models to enhance the accuracy of inflation rate forecasting, specifically comparing the performance of the Extreme Gradient Boosting (XGBoost) algorithm with the Random Forest (RF) algorithm.
- Random Forest is an ensemble learning technique that constructs multiple decision trees and aggregates their predictions, making it robust to overfitting and effective .

Fig1. Market Tremors

MATERIALS AND METHODS



Data Collection → Data Pre-processing → Splitting data into training and testing data → Models training (XG-Boost and DT) → Model validation → Performance comparison of XG-Boost with RF

RESULTS

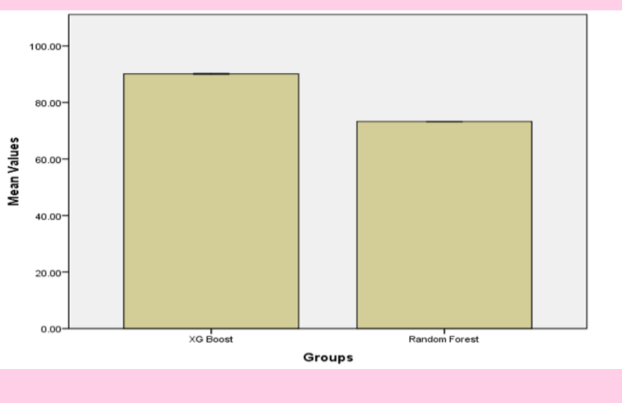


Fig 2 . XGBoost and RF

Table 1 . Standard deviation and mean value of XGBoost and RF

	Algorithm	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	XG-Boost	10	90.115	0.084	0.02659
	RF	10	73.240	0.00527	0.0167

DISCUSSION AND CONCLUSION

- According to the study's findings, XG-Boost predicts inflation rates better than the Random Forest algorithm. Random Forest's accuracy was 78.08%
- whereas XG-Boost's precision was 90.11%. These findings suggest that XG-Boost provides a more accurate and dependable way to predict inflation trends & p=0.45.
- So, we conclude that XG-Boost outperforms random forest algorithm

BIBLIOGRAPHY

- Beer, Christian, Robert Ferstl, and Bernhard Graf. 2025. "Improving Disaggregated Short-Term Food Inflation Forecasts with Webscraped Data." 262. Working Paper. https://hdl.handle.net/10419/312869.
- "Global Inflation Forecasting and Uncertainty Assessment: Comparing ARIMA with Advanced Machine Learning." 2025. Journal of Radiation Research and Applied Sciences 18 (2): 101402.
- Vilenko, Maya. 2025. "BiHRNN -- Bi-Directional Hierarchical Recurrent Neural Network for Inflation Forecasting." http://arxiv.org/abs/2503.01893.

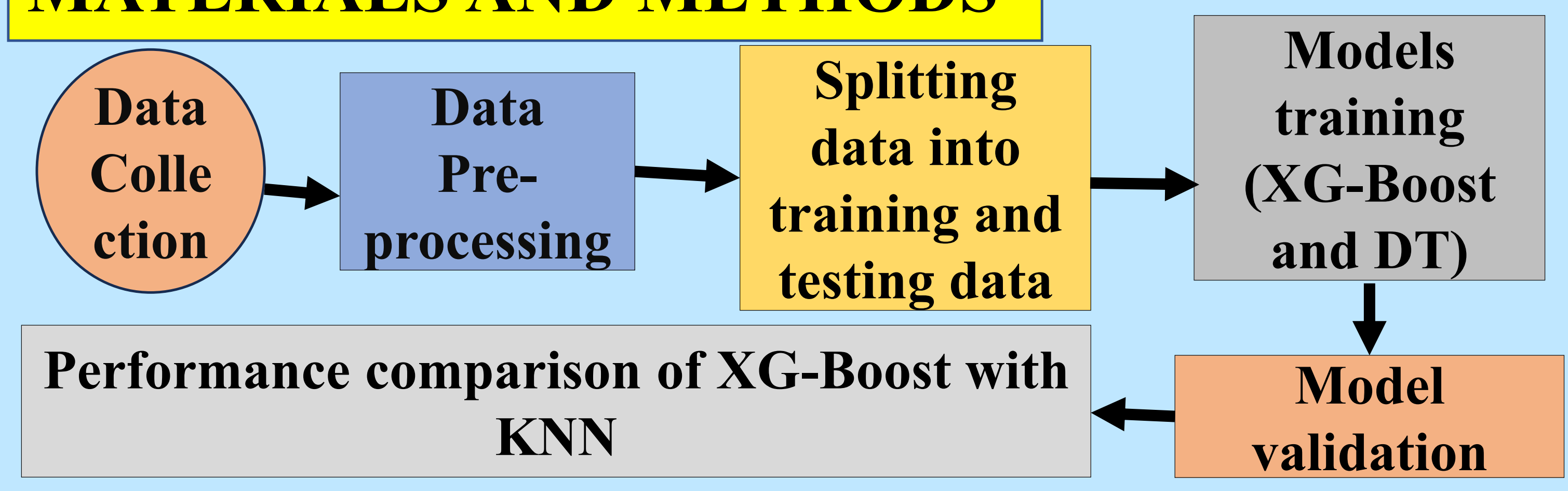
Advanced Accuracy Prediction in Inflation Rate Using Extreme Gradient Boosting Algorithm in Comparison with K-Nearest Neighbors Algorithm

INTRODUCTION

- The research explores the use of machine learning techniques to improve the accuracy of inflation rate prediction, focusing on a comparative analysis between XGBoost algorithm and the K-Nearest Neighbors (KNN) algorithm.
- Extreme Gradient Boosting, in contrast, builds models iteratively, where each new tree corrects the errors of the previous ones.

Fig1. Dollar`s Decent: shrinking purchasing power

MATERIALS AND METHODS



Data Collection → Data Pre-processing → Splitting data into training and testing data → Models training (XG-Boost and DT) → Model validation → Performance comparison of XG-Boost with KNN

RESULTS

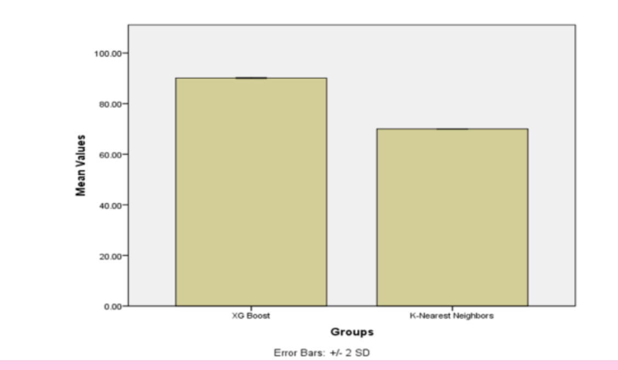


Fig 2 . XGBoost and KNN

Table 1 . Standard deviation and mean value of XGBoost and KNN

	Algorithm	N	Mean	Std. Deviation	Std. Error Mean
Accuracy	XG-Boost	10	90.115	0.084	0.02659
	KNN	10	70.02	0.001	0.01

DISCUSSION AND CONCLUSION

- According to the study's findings, XGBoost predicts inflation rates more accurately than the KNN algorithm. KNN trailed behind with an accuracy of 70.00%
- whereas XGBoost obtained a greater accuracy of 90.11% & p=0.45. These results demonstrate that the XGBoost represents a more precise and trustworthy forecasting method.

BIBLIOGRAPHY

- Beer, Christian, Robert Ferstl, and Bernhard Graf. 2025. "Improving Disaggregated Short-Term Food Inflation Forecasts with Webscraped Data." 262. Working Paper.
- Vilenko, Maya. 2025. "BiHRNN -- Bi-Directional Hierarchical Recurrent Neural Network for Inflation Forecasting." http://arxiv.org/abs/2503.01893.
- "Global Inflation Forecasting and Uncertainty Assessment: Comparing ARIMA with Advanced Machine Learning." 2025. Journal of Radiation Research and Applied Sciences 18 (2): 101402.